



PES University, Bengaluru
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END SEMESTER ASSESSMENT (ESA) - Dec 2024

UE21CS352B - Object Oriented Analysis & Design using Java

Total Marks : 100.0

1.a. Draw a use case diagram for railway reservation system which has all basic functionalities similar to IRCTC portal by identifying at least 5 usecases. (6.0 Marks)

1.b. With respect to class diagram explain the following terms with an example:
Composition
Aggregation
Association
Qualified Association
Generalization
Recursive relation (12.0 Marks)

1.c. For the below given scenario draw an activity diagram

In a Flight Reservation System, a customer books a flight through an online platform. The booking process involves several subsystems and conditional flows like checking for flights, selecting seats, processing payment, and notifying customers.

The customer searches for available flights based on origin, destination, and date. The customer enters personal details, including name, contact info, and travel document details to book a ticket. The customer decides whether to add services like extra baggage, in-flight meals, or travel insurance. This step is optional. A confirmation email or SMS with the booking details is sent to the customer. System should also handle payment failures. (7.0 Marks)

2.a. Design a class diagram for a system to manage online shopping based on the information given below. In an online shopping system, several complex subsystems need to work together to place an order. These subsystems can include:

1. Inventory System – Checks if the items are in stock. (Handle different types of items like perishable and non-perishable)
2. Payment System – Processes the payment.
3. Shipping System – Handles shipping details.
4. Notification System – Sends order confirmation emails to the customer.

Ensure that the constructors and destructors are appropriately used for each class. Demonstrate overloading of methods to display details of both perishable and non perishable items. (7.0 Marks)

2.b. Answer the following:

1. What is the need for interfaces in java.
2. Write a Java program that demonstrates the use of an abstract class. The program should include the following:
 - An abstract class Shape with an abstract method area() to calculate the area of a shape.
 - Two subclasses Circle and Rectangle that extend Shape. Each subclass should implement the area() method to calculate the area of a circle and a rectangle respectively.
 - In the main method, create objects of both Circle and Rectangle and calculate the area for both shapes.

(6.0 Marks)

2.c. Write a Java program to demonstrate the use of the super keyword in the following scenarios:

1. Accessing a superclass constructor from a subclass.
2. Accessing a superclass method from a subclass.

(6.0 Marks)

2.d. Answer the following

i) What is the output of the below given program?

```
public class Example {  
    static int count = 0;  
    public Example() {  
count++;  
    }  
    public static void displayCount() {  
        System.out.println("Count: " + count);  
    }  
    public static void main(String[] args) {  
        Example obj1 = new Example();  
        Example obj2 = new Example();  
        Example.displayCount();  
    }  
}
```

ii) State the difference between static and non-static members in Java.

(6.0 Marks)

3.a. A library management system needs to keep track of books borrowed by members. Each member can borrow multiple books, and the system must calculate the total number of books borrowed by all members in the library. Using the Information Expert principle, identify which class should be responsible for calculating the total number of borrowed books. Justify your choice and design the solution.

(6.0 Marks)

3.b. Suppose you have a `Payment_Processor` class that handles credit card payments and sends email notifications.

Which SOLID principles does it violate and why?

How would you redesign it?

(6.0 Marks)

3.c. Explain lazy and early implementation of Singleton Design pattern using Java. Highlight its advantages and disadvantages

(6.0 Marks)

3.d. Explain the builder pattern with one relevant example

(7.0 Marks)

4.a. Differentiate between pattern and anti-pattern

(6.0 Marks)

4.b. What is a proxy pattern? Which are the three main variations of proxy pattern. (6.0 Marks)

4.c. What are anti patterns? State the solution for Blob, Lava Flow, Vendor lockin and Analysis-Paralysis antipatterns. (5.0 Marks)

4.d. Explain the Facade Design Pattern with a suitable real-world example. Discuss how it helps simplify client interaction with complex systems. Provide a brief code example to illustrate your explanation. (8.0 Marks)